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Unit – 4 : Session – 3 : SLO – 1 | Custom Higher-Order Functions

■ I. Fill in the Blanks

1. lambda — a lambda function in Python is a short anonymous function.
2. function — a higher-order function can take another function as an argument (or return one).

■ II. Multiple Choice

1. `list(filter(lambda x: x % 2 == 0, [1, 2, 3, 4]))` → [2, 4] — filter keeps only the even numbers.

■ III. Example — Custom Higher-Order Function

- `def apply_twice(func, value): return func(func(value))`
- `apply_twice(lambda x: x * 3, 2)` # → 18
- Higher-order functions treat functions as first-class values: they can be passed, returned and stored like any other object.